One Configuration, Tuned per Dataset

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The configuration. Inside and outside edges get two separate colors (their own weights in the GNN), trained with GLASS's recipe on GLASS's splits. With copies, every shared node is replaced by one copy per subgraph.

What we ran, per dataset:

1. An Optuna search over aggregation, pooling, hidden size, layers, lr, dropouts, batch size, weight decay, norm, jumping knowledge, head, and (synthetic only) edge counts. Each trial is 3 seeds, scored by mean validation. Trials: hpo_neuro 80, hpo_metab and em_user 160, the rest 300. Weak trials were stopped after seed 0.
2. The 3 best trials on 10 seeds. The one with the best 10-seed validation is reported. Test was never used to choose.

The table shows test micro-F1 (%), mean ± std over 10 seeds; bold is the best in each row.

- *One config:* the search above. The real datasets ran without copies, because their copies graphs did not fit in GPU memory then. The synthetic datasets ran with copies.
- *Copies, free coloring:* the real datasets searched again with copies. The search also picked the coloring: inside / outside, inside split by copy, or inside / outside plus clique. Attention was left out.
- *Copies, one coloring:* inside / outside with copies everywhere. On ppi_bp, hpo_neuro and em_user, it reuses the free-coloring winner's settings with only the coloring changed.
- *Copies, more regularization:* the 3 free-coloring finalists of ppi_bp and em_user, each with more dropout (+0.2), more weight decay (×10, at least 10^{-3}), or both. The one with the best validation is reported.
- *GLASS and GPEN:* published numbers from the GPEN paper.

	ours				published	
			copies			
dataset	one config	free coloring	one coloring	more regularization	GLASS	GPEN
ppi_bp	55.3 ±0.9	58.6 ±2.2	60.6 ±2.3	60.4 ±1.1	61.9 ±2.2	**64.4** ±0.9
hpo_metab	60.5 ±2.8	**65.0** ±2.1	**65.0** ±2.1	–	61.4 ±1.6	63.9 ±0.9
hpo_neuro	64.6 ±0.7	68.4 ±0.5	67.6 ±1.1	–	68.5 ±1.6	**69.1** ±0.6
em_user	82.9 ±7.3	81.2 ±3.2	69.0 ±6.7	80.2 ±3.7	88.8 ±1.9	**91.2** ±1.3
density	**98.9** ±1.3	–	**98.9** ±1.3	–	93.0 ±2.8	95.6 ±1.6
cut_ratio	**95.2** ±2.0	–	**95.2** ±2.0	–	93.5 ±1.9	93.6 ±1.6
coreness	86.7 ±5.2	–	86.7 ±5.2	–	84.0 ±2.8	**87.6** ±1.9
component	**100.0** ±0.0	–	**100.0** ±0.0	–	**100.0** ±0.0	**100.0** ±0.0

Notes.

- em_user colorings. Copies, free coloring: inside / outside plus the clique, with copies. Copies, one coloring: inside / outside with copies.
- Why ppi_bp and em_user trail GLASS and GPEN, two guesses:
  1. Overfitting. Train F1 is about 99.8 on both em_user and ppi_bp, against 89 on hpo_metab, where we beat the baselines.
  2. Noisy validation. Both have small validation sets (em_user has about 65 subgraphs), so the search picks winners by luck. ppi_bp's winner scores 69.6 on validation but 58.6 on test, and on em_user validation and test barely track each other (0.22 correlation).